Team BabyBouncers

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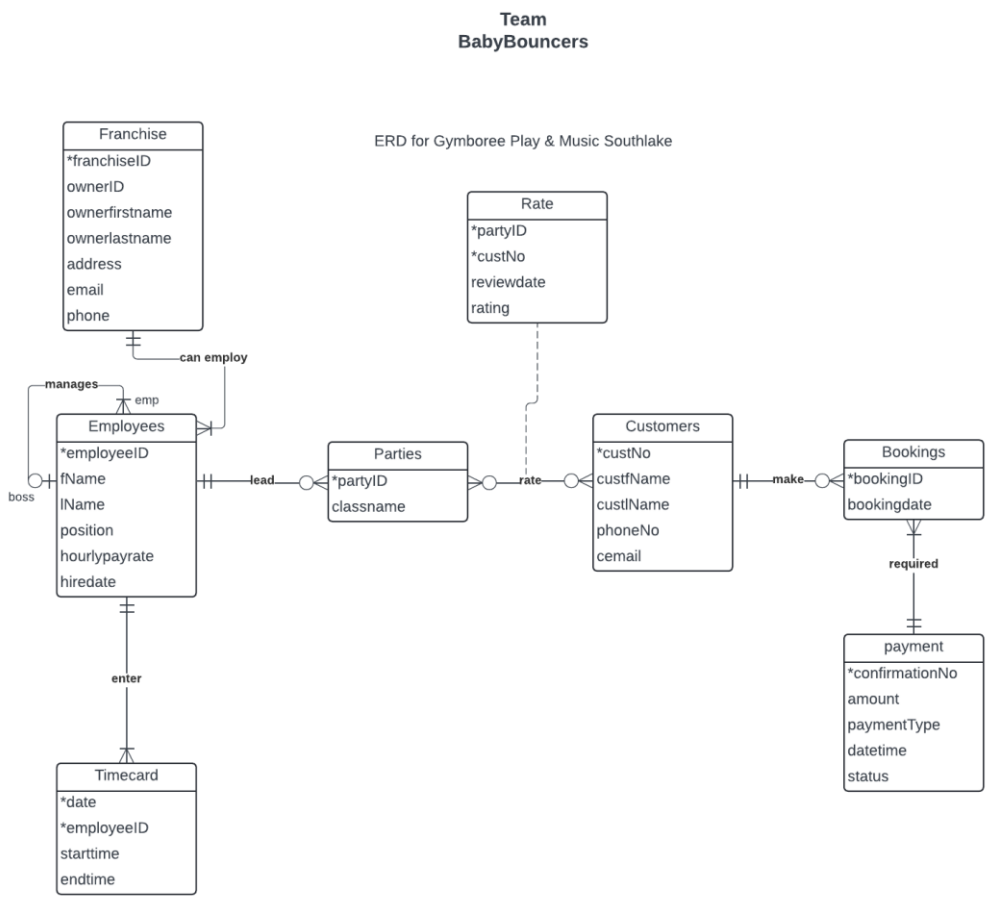
Part I - Business Description

BabyBouncers specializes in providing a special private party experience tailored for children, featuring a variety of child-friendly activities designed to create an unforgettable event. Our personalized service is dedicated to crafting a unique event that caters to the individual preferences and needs of each child attending the party. Our private party setting ensures an intimate and secure environment where children can enjoy activities comfortably and safely. We offer a diverse selection of entertaining and engaging activities, likely allowing children to remain captivated and delighted throughout the celebration. Our focus on child-friendly entertainment ensures that every aspect of the party is specifically designed with young guests in mind, creating a joyful experience for everyone involved.

Part II - ER Diagram



Part III - Relational Data Model

employees (**employeeID**, fName, lName, position, hourlypayrate, hiredate, */FranchiseID/, /bossID/*)
franchise (**franchiseID**, ownerID, ownerfirstname, ownerlastname, address, email, phone)
timecard (**date**, ***/employeeID/***, starttime, endtime)
parties (**partyID**, classname, */employeeID/*)
customers (**custNo**, custfName, custlName, phoneNo, cemail)
rate (***/partyID/***, ***/custNo/***, reviewdate, rating)
bookings (**bookingID**, bookingdate, */custNo/,/confirmationNo/*)
payments (**confirmationNo**, amount, paymentType, datetime, status)

Part IV - Data Dictionary

Column Name	Data Type	Field Size	Required	Definition
address	VARCHAR	60	YES	Franchise Address
amount	SMALLINT	-	YES	Payment amount
bookingdate	DATE	-	YES	Booking date
booking_id	INT	-	YES	Booking ID
bossID	SMALLINT	-	YES	Employee's Boss ID
cemail	VARCHAR	40	YES	Customer's email
classname	VARCHAR	15	YES	Party's class name
confirmationNo	INT	-	YES	Payment confirmation number
custfName	VARCHAR	20	YES	Customer first name
custlName	VARCHAR	20	YES	Customer last name
custNo	SMALLINT	-	YES	Customer number
date	DATE	-	YES	Date on employee's timecard
datetime	TIMESTAMP	-	YES	Payment date & time
endtime	TIME	-	YES	End time on employee's timecard
email	VARCHAR	40	YES	Franchise owner's email
employeeID	SMALLINT	-	YES	Employee's ID
fName	VARCHAR	20	YES	Employee's first name
franchiseID	SMALLINT	-	YES	Franchise ID
hiredate	DATE	-	YES	Employee's hire date

hourlypayrate	SMALLINT	-	YES	Employee's hourly pay rate
lName	VARCHAR	20	YES	Employee's last name
ownerfirstname	VARCHAR	20	YES	Franchise owner's first name
ownerID	SMALLINT	-	YES	Franchise owner's ID
ownerlastname	VARCHAR	20	YES	Franchise owner's last name
partyID	SMALLINT	-	YES	Party ID
paymentType	VARCHAR	10	YES	Payment type
phone	VARCHAR	20	YES	Franchise owner's phone number
phoneNo	VARCHAR	20	YES	Customer's phone number
position	VARCHAR	11	YES	Employee's position
rating	SMALLINT	-	YES	Customer's rating
reviewdate	DATE	-	YES	Review date on customer rating
starttime	TIME	-	YES	Start time on employee's timecard
status	VARCHAR	10	YES	Payment status

Part V – Query Matrix, SQL Queries, Trigger, and View

A. Query Matrix

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
1. More Than 2 Tables Joined	X									
2. Subquery		X								
3. COUNT with DISTINCT			X							
4. GROUP BY with HAVING				X						
5. ORDER BY					X					
6. Aggregation Function		X		X		X	X			
7. Query from View							X			
8. DATE Calculation					X	X		X		X
9. Self-joining Query (Recursive)									X	
10. Calculated field with alias										X

B. SQL Queries

1.List the employees who instruct the ‘Music 1’ parties

```
select e.employeeID, e.fName, e.lName, e.position, p.partyID, p.classname, r.rating

from employees e, parties p, rate r

where e.employeeID=p.employeeID

and p.partyID = r.partyID

and e.position='instructor'

and p.classname='Music 1';
```

2.What classes have a rating greater than the average rating for the classes?

```
select p.classname, r.partyID, r.rating

from parties p, rate r

where p.partyID= r.partyID

and r.rating > (select avg (r.rating) from rate r);
```

3.How many different classes does Mauricio teach at parties?

```
select count(distinct p.classname), e.employeeID, e.fName, e.lName, e.position
from parties p, employees e
where p.employeeID= e. employeeID
and e.fName= 'Mauricio'
group by e.employeeID;
```

4.For each class with more than 1 rating, what is the average rating for the class and the sum of the ratings?

```
select round(avg(r.rating), 2) as averagerate, sum(r.rating) as ratesum, p.classname
from rate r, parties p
where r.partyID = p.partyID
group by p.classname
having count(r.rating)>1;
```

5.Order the employee timecards by employee ID (ascending).

```
select employeeID, starttime, endtime, date
from timecard
order by employeeID asc;
```

6.Find the sum of all the payments made using ‘credit’ for the month of June.

```
select sum(amount), extract(month from datetime) as month
from payments
where paymentType = 'credit' and extract(month from datetime) = 6
group by extract(month from datetime);
```

7.How many of the employees are instructors? Note: CREATE VIEW of employees table and remove hourlypayrate, hiredate, franchiseid, and bossid to have 'manager' pull this data for 'owner'?

```
create view employeetableforowner as
select employeeID, fName, lName, position
from team3_database.employees;

select * from employeetableforowner;
select count(employeeID)
from employeetableforowner
where position='instructor';
```

8.Separate the Date and Time from the ‘Datetime’ column in the payments table.

```
select cast(datetime as time) as Time, cast(datetime as date) as Date
```

```
from payments;
```

9. List all of the front-end employees who are managed by June (first name, last name, and the date they were hired).

```
select e.fName, e.lName, e.hiredate
from employees e, employees b
where b.employeeID= e.bossID
and e.position= 'frontend'
and b.fName= 'June';
```

10.On June 4th there was a special going on at BabyBouncers for 20% off party bookings. The discount was not taken off customers' payments. Correct the payments to display the correct amounts (also include the confirmation number and the customer number)

```
SELECT b.confirmationNo, b.custNo, p.amount *0.80 AS 'correctamount'
FROM payments p, bookings b
WHERE p.datetime like '2024-06-04%';
```

C. Trigger

```
CREATE TRIGGER timecard_update
AFTER UPDATE ON timecard
FOR EACH ROW
        INSERT INTO timecard_log
        VALUES
(old.date,old.employeeID,old.starttime,new.starttime,old.endtime,new.endtime,current_ti
mestamp());
```

DML statement to fire the trigger:

```
UPDATE timecard
SET date='2024-06-15',employeeID=12,starttime = '10:00:00', endtime = '18:30:00'
WHERE date='2024-06-15' AND employeeID=12;
```

D. View

```
CREATE VIEW employeetableforowner AS

SELECT employeeID, fName, lName, position

FROM team3_database.employees;
```